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import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Step 1: Simulated microarray data (genes as features)
np.random.seed(0)
X = np.random.rand(100, 50)   # 100 samples, 50 genes
y = np.random.randint(0, 2, 100)  # Disease (1) / Normal (0)

# Step 2: Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2)

# Step 3: Model training
model = RandomForestClassifier()
model.fit(X_train, y_train)

# Step 4: Prediction
y_pred = model.predict(X_test)

# Step 5: Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))
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- Accuracy: 0.5